`main (27).pdf` - Corrections to apply

1) In Section 6.3 (page 10), where it says:

> "Second, on the SFT vs. DPO pairs, DPO wins every consistent verdict across all four betas (0 SFT wins in 25 consistent pair-prompt verdicts), with zero losses."

It should say:

> "Second, on the SFT vs. DPO pairs, DPO wins every consistent verdict on the three in-range betas (18 of 18 across the SFT-vs-DPO-b01, b02 and b03 pairs, zero losses). On the SFT vs. DPO-b005 pair the signal weakens (7 DPO wins, 2 SFT wins on consistent verdicts), the first hint of the β=0.05 cliff that perplexity flags more sharply in Section 6.2."

Why: Table 8 (next page) shows 2 SFT wins on the b005 leg, and the "25 consistent verdicts" number conflates DPO-wins (25) with consistent verdicts (27 = 25 DPO + 2 SFT). §6.5's "18 of 18 across the three in-range betas" phrasing is correct; §6.3 should match it.

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2) In Section 6.5 (page 12), where it says:

> "The deliverable policy is DPO-b01 (β = 0.10, learning rate 5×10^{-5}): it beats the base 18-0 with 90% judge agreement, beats SFT 8-0 in consistent verdicts, and shows no fluency collapse or structural regression relative to SFT."

It should say:

> "The deliverable policy is DPO-b01 (β = 0.10, learning rate 5×10^{-5}): it beats the base 18-0 with 90% judge agreement, beats SFT 8-0 in consistent verdicts, and shows no fluency collapse. Structural completeness is comparable to SFT (1/20 vs 3/20 all-four-section

> generations; 2.50 vs 2.80 average sections per generation): a small decrement that is a known cost of restoring `repetition_penalty=1.3`, which occasionally triggers an early EOS before the model reaches `## Requirements`. Section 7 documents this trade-off."

Why: Verified directly from `data/processed/ma2/eval_generations/*.jsonl`. SFT has 3/20 all-four-section generations and 2.80 avg sections; b01 has 1/20 and 2.50 avg sections. The "no structural regression" claim contradicts Table 9 on the same page.

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3) In the notebook cell `ma2s66` (Section 6.6), where it says:

> "All four DPO models produce all four required Markdown headings on every one of the 20 evaluation generations."

It should say:

> "Structural completeness varies across models. SFT and DPO-b03 produce all four required Markdown headings on 3 of 20 generations; DPO-b01 on 1 of 20; DPO-b02 and DPO-b005 on 0 of 20. Average required-heading counts per generation are 2.80 (SFT), 2.85 (b03), 2.50 (b01), 2.45 (b02), 1.35 (b005). The shortfall is the rep_penalty trade-off: restoring `repetition_penalty=1.3` (Section 6.2 and Section 7.3.1) eliminates the token-loop catastrophes but occasionally triggers an early EOS before the model reaches `## Requirements`."

Why: Verified by direct count over `data/processed/ma2/eval_generations/*.jsonl`. The notebook prose contradicts the report's Table 9.

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4) In the notebook cell `ma2s71` (Section 7.1), where it says:

> "all four DPO models produce the four required Markdown sections on every one of the 20 evaluation generations."

It should say:

> "structural completeness lands within a narrow band of SFT but is generally lower than the 4/20 all-four-section count one would want for production: 3/20 (SFT), 3/20 (b03), 1/20 (b01), 0/20 (b02), 0/20 (b005). The shortfall is the known cost of the `repetition_penalty=1.3` kwarg restoration documented in §7.3.1."

Why: Same defect as #3, in the limitations chapter rather than the behavioural-stats chapter. Both must be fixed together.

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5) In Section 6.5 (page 12), where it says:

> "The β = 0.05 probe breaks this pattern on every axis: a 47% perplexity jump, a 40% drop in mean output length, and structural statistics worse than SFT, combined with a win-rate signal that becomes noise (DPO-b005 vs. DPO-b01 is 2-1 with 85% inconsistency). The alignment-quality peak sits at β = 0.10."

It should say:

> "The β = 0.05 probe breaks this pattern on every axis: a 47% perplexity jump, a 40% drop in mean output length, and structural statistics worse than SFT, combined with a win-rate signal at the noise floor (DPO-b005 vs. DPO-b01 is 2-1 with 85% inconsistency). Among the tested configurations, β = 0.10 is the strongest candidate by every measurable axis (eval reward accuracy peak at 0.715; consistent close-pair win-rate; proportionate perplexity rise); the low close-pair agreement rates documented in Section 7 prevent a statistically robust ranking between the three in-range DPO variants."

Why: The same close-pair agreement rates (15-40%) that the report acknowledges in §7 prevent strict ordering between b01, b02 and b03 at N=20. The categorical "peak sits at β = 0.10" overstates what the data can support.

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6) In Section 7.2 (page 13), the closing paragraph that asserts the

β = 0.10 peak. Where it says (search for the categorical "alignment-quality peak" phrasing):

> "...the alignment-quality peak sits at β = 0.10..."

It should say:

> "...β = 0.10 is the strongest candidate among the tested configurations on every measurable axis, although the close-pair AB-BA agreement rates documented above prevent a statistically robust ranking between the three in-range DPO variants..."

Why: Same softening as #5, second occurrence.

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7) In Section 8 Conclusion (page 15), where it says:

> "A four-point beta sensitivity probe located the alignment-quality peak at β = 0.10: perplexity and win-rate both degrade below that value, with β = 0.05 producing a 47% perplexity cliff and a 40% drop in mean output length."

It should say:

> "A four-point beta sensitivity probe identifies β = 0.10 as the strongest candidate among the tested configurations: perplexity and win-rate both degrade below that value, with β = 0.05 producing a 47% perplexity cliff and a 40% drop in mean output length. The close-pair AB-BA agreement rates documented in Section 7.2 prevent a

> statistically robust ranking between the three in-range DPO variants; β = 0.10 is the recommended deliverable on best-available-evidence grounds rather than as a proven optimum."

Why: Same softening as #5, third occurrence (the headline phrasing in the Conclusion).

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8) In Table 4 (page 6), the `ood-03` column currently shows:

Model	ood-03
gemma-4 (control)	(blank)
nemotron	(blank)
granite-3.3	(blank)
smollm3	×
qwen3.5	×
ministral	(blank)

It should show:

Model	ood-03
gemma-4 (control)	✓
nemotron	✓
granite-3.3	✓
smollm3	×
qwen3.5	×
ministral	✓

Why: Verified via `pdftotext` extraction that the cells are genuinely blank in the rendered PDF. The semantic ("blank = caught the collapse") is inferable but ambiguous; a reader scanning quickly may assume "blank" means "not tested" or "data unavailable". Explicit checkmarks resolve the ambiguity. The cells that pass the disqualifier should display ✓ rather than nothing.

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9) In Section 6.5 (page 12), where the deliverable is named without

explaining the decision rule. Where it says:

> "The deliverable policy is DPO-b01 ($\beta = 0.10$, learning rate 5×10^{-5}): it beats the base 18-0 with 90% judge agreement, beats SFT 8-0 in consistent verdicts, and shows no fluency collapse or structural regression relative to SFT."

(this paragraph is also targeted by item 2 above; the rule clarification should be applied to the same paragraph after the structural-regression correction is in place)

A new sentence should be inserted before "The deliverable policy is DPO-b01..." to articulate the decision rule:

> "Among the in-range betas, perplexity is lowest at $\beta = 0.30$ (4.80) and rises monotonically as β decreases, while reward accuracy peaks at $\beta = 0.10$ (0.715) and consistent close-pair verdicts sort lower-beta-preferred. Selecting the deliverable prioritises the alignment metrics (preference accuracy and pairwise win-rate) over absolute fit to the SFT distribution, provided perplexity remains within a proportionate range of SFT (which it does at $\beta = 0.10$: 6.01 vs SFT's 4.54, a 32% rise)."

Why: Verified by full report text scan: the report names DPO-b01 as the deliverable but never states why over $\beta = 0.20$, which has lower perplexity (5.03 vs 6.01). The selection rule is implicit. A reader's natural question is left unanswered. The added sentence makes the decision rule explicit.

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10) In Section 5.4 (page 8), where it says:

> "Five epochs over 1,302 triples was determined empirically: fewer epochs left reward margins near zero; more caused the train reward accuracy to saturate at 1.00 with no corresponding deployment-side benefit (see Section 7)."

It should say:

> "Five epochs over 1,302 triples was determined empirically: fewer epochs left reward margins near zero; at five epochs the training reward accuracy already saturates at 1.00 across all four betas, so additional epochs would not improve the training-side signal and risk further fluency degradation (see Section 7)."

Why: Verified from `outputs/ma2-360m-dpo-*/log_history.json`: train_rwd_acc reaches 1.00 at 5 epochs across all four betas. The "more caused saturate" comparative implies a 6+ epoch experiment was run; none was. The saturation is already at the chosen setting.

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11) In Section 6.1 (page 9), where it says:

> "All six models are evaluated with greedy decoding and repetition_penalty=1.3, matching the Mini-Assignment 1 inference helper."

It should say:

> "All six models are evaluated with greedy decoding, `repetition_penalty=1.3`, and `max_new_tokens=800`, matching the Mini-Assignment 1 inference helper for the first two kwargs. The Section 3.4 sanity-check generations use the same decoder configuration with `max_new_tokens=400`; the Section 4.2 candidate sampling uses `temperature=0.9`, `top_p=0.95`, `num_return_sequences=4` and `max_new_tokens=500`."

Why: Verified by full report text scan: `max_new_tokens` is not stated for §3.4 sanity-check generations or §6.1 evaluation generation, though both are present and important for reproducibility. The notebook cells `ma2s35code` (Section 3.5 sanity check) use 400, `ma2s55code` (Section 5.5 sanity check) use 400, `ma2s62code` (Section 6.2 eval generation) use 800. The consolidated sentence above replaces the existing one-liner with a complete generation-kwarg summary covering the three stochastic/greedy decoding stages.

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12) In Section 7.1 (page 12), where it says:

> "DPO-β=0.20 on the cloud-engineer prompt ind-07 emitted 42 consecutive repetitions of the token SQS as bullet items in the Required Skills section; DPO-β=0.30 on the same prompt emitted 29 consecutive ECS repetitions."

It should say:

> "DPO-β=0.20 on the cloud-engineer prompt ind-07 emitted a Required Skills bullet in which the AWS service token SQS repeated 171 times in a comma-separated list before EOS; DPO-β=0.30 on the same prompt emitted an escalating-bullet degeneration in which the ECS token grew across consecutive bullet items ('ECS Pods', 'ECS ECS Pods', 'ECS ECS ECS Pods', ...) for over 25 lines."

Why: The "42" and "29" were sliding-window n-gram repeat counts from the original diagnostic scanner, not bullet-item counts. The b02 case manifests as a comma-separated list within a single bullet (171 raw SQS tokens); only the b03 case presents as escalating bullet items. The qualitative finding is unchanged; the geometric description matches the actual data.

Summary

#	Location	Type	Priority
1	Report §6.3	Factual count error, contradicts Table 8	**high**

#	Location	Type	Priority
2	Report §6.5	Overclaim, contradicts Table 9 on same page	**high**
3	Notebook `ma2s66`	All-sections claim contradicts on-disk data	**high**
4	Notebook `ma2s71`	Same defect as #3 in the limitations chapter	**high**
5	Report §6.5	"Peak sits at" overconfident given §7 caveats	medium
6	Report §7.2	Same overconfidence in beta-sensitivity closing	medium
7	Report §8	Same overconfidence in Conclusion	medium
8	Report Table 4	ood-03 blank cells, fill with ✓ for passing models	medium
9	Report §6.5	Missing decision rule (b01 over b02)	medium
10	Report §5.4	"More epochs caused saturate" implies untested	medium-low
11	Report §6.1	`max_new_tokens` missing for §3.4 and §6.1 generation	low
12	Report §7.1	"42 SQS as bullet items" geometry off	low

Items 1-4 are factual contradictions visible to any careful reader of the report (or report + notebook); they are the only items that produce an internal inconsistency on inspection. Items 5-7 are the same overconfidence pattern in three locations and can be fixed together. Items 8, 9 are visible content gaps (a blank table column, a missing decision rule for the deliverable). Item 10 implies an experiment that was not run. Item 11 is a reproducibility kwarg gap. Item 12 is a descriptive nuance where the underlying finding is unchanged.